

AI-Based Network Intrusion Detection System

Final Project Report

Project Status: COMPLETED

Development Phases: 12

Repository: yuvarajcybersec/Ai-Network-Intrusion-Detection-System

Final Phase: Phase 12 – Final Repository Validation & Project Review

1. Executive Summary

The AI-Based Network Intrusion Detection System is a modular defensive cybersecurity project designed to monitor network traffic, extract packet-level features, classify traffic using machine learning, generate alerts for suspicious activity, and provide monitoring and statistical analysis.

The project was developed progressively through twelve structured phases covering environment preparation, packet capture, feature extraction, dataset preparation, feature engineering, machine learning, real-time detection, alert generation, monitoring, end-to-end integration, performance evaluation, deployment reproducibility, automated testing, and final repository validation.

The final validation confirmed that the project is operational as a complete prototype. The automated test suite completed successfully with 26 passing tests. Python source compilation passed, the trained Random Forest model was successfully loaded and validated, the feature dataset contained no missing values, evaluation artifacts were present, repository integrity checks passed, and the Git working tree was clean and synchronized with the remote repository.

The project is therefore ready for final academic or internship submission.

2. Project Objectives

The major objectives were:

- Capture network traffic using Python and Scapy.
 - Extract structured packet-level network features.
 - Prepare labeled traffic datasets.
 - Engineer machine-learning features.
 - Train a supervised Random Forest classifier.
 - Perform real-time traffic classification.
 - Generate alerts for suspicious traffic.
 - Monitor and analyze generated alerts.
 - Integrate the IDS components into an operational workflow.
 - Evaluate model performance.
 - Validate deployment reproducibility.
 - Implement automated regression testing.
 - Perform a final repository and release-readiness review.
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3. System Architecture

The project follows a modular IDS pipeline:

“text Network Interface | v Packet Capture | v Feature Extraction | v Labeled Traffic Dataset | v Feature Engineering | v Random Forest Model | v Real-Time Detection | v Alert Manager | v Alert Monitoring & Statistics